

the bottom of the container (tank). IC 74HC147 has nine active-low inputs and converts the active input into active-low BCD output. The input L-9 has the highest priority.

The outputs of IC1 (A, B, C and D) are fed to IC2 via transistors T1 through T4. This logic inverter is used to convert the active-low output of IC1 into active-high for IC2. The BCD code received by IC2 is shown on 7-segment display LTS543. Resistors R18

through R24 limit the current through the display.

When the tank is empty, all the inputs of IC1 remain high. As a result, its output also remains high, making all the inputs of IC2 low. Display LT5543 at this stage shows '0', which means the tank is empty. Similarly, when the water level reaches L-1 position, the display shows '1', and when the water level reaches L-8 position, the display shows '8'. Finally, when

the tank is full, all the inputs of IC1 become low and its output goes low to make all the inputs of IC2 high. Display LTS543 now shows '9,' which means the tank is full.

Assemble the circuit on a general-purpose PCB and enclose in a box. Mount 7-segment LTS543 on the front panel of the box. For sensors L-1 through L-9 and ground, use corrosion-free conductive-metal (stainless-steel) strips. ●

VERSATILE PROBE

■ RAJU R. BADDI

You can use this versatile probe for continuity testing and identification of transistor type and transformer windings. The n-side or p-side of a transistor can be identified quickly in one go. You can make two contacts with the probe in one hand while the other hand is free.

Fig. 1 shows the circuit of the probe. The operation of the circuit is simple. It is driven by an alternating current flowing through two LEDs (LED1 and LED2). So one LED corre-

sponds to forward direction of current flow, while the other shows reverse direction of current flow. This helps to detect orientation of the p-n junction with respect to the probes. The LEDs can be arranged near the probes to glow either for the p-side or the n-side as per your choice.

The frequency is determined by capacitor C1 and preset VR1 connected between gates G1 and G2. The frequency can be varied using preset VR1. Higher frequency results in more sensitivity to inductive reactance. The preset is trimmed so that when the probes are shorted, both the LEDs glow equally.

Fig. 2 shows the probe arrangement for testing. Most of the battery power is consumed only when the LEDs glow. The probes have been constructed to provide a good grip on the components under testing. One probe's tip has been

widened. (Drop the empty refill of a ball-pen from some height to remove the ball, then insert a sharp needle or something similar into the tip. Slowly push the needle inside and widen the tip so that a component lead can be

